INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
(School of University of Engineering and Management, Kolkata)

End Semester Examination, April - May, 2026

Degree: **B.C.A.** Semester: **Second**

Stream: **B.C.A.**

Paper Code: **BCAMD201**

Paper Name: **Mathematics for Computing**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. For any two events A_1 and A_2 where $A_1 \subset A_2$, state the correct relationship involving their probabilities. **2**

2. In a single cast with two dice, the chance of one die turns up 3 and the other shows 4. Find the probability of this event occurring. **2**

3. A bag contains 5 brown balls and 4 white balls. A man pulls out two balls from the bag. Find the probability that both balls drawn are of the same colour. **2**

4. At an examination, the probabilities that a student gets first class, second class, and third class are $\frac{1}{10}$, $\frac{2}{5}$, and $\frac{1}{5}$ respectively. Find the probability that the student fails. **2**

5. Given that $P(B) = \frac{3}{4}$, $P(\bar{A} \cap B \cap \bar{C}) = \frac{1}{3}$, $P(A \cap B \cap \bar{C}) = \frac{1}{3}$, find the value of $P(B \cap C)$. **2**

6. Find the arithmetic mean (A.M.) of the following frequency distribution: **2**

x	:	6	5	2	8
Frequency:		4	3	1	2

7. Find the median of the following distribution: 10, 13, 9, 7, 37, 16, 27, 32. **2**

8. If two sets of observations $\{x_i\}$ and $\{y_i\}$ are related by the equation **2**

$x_i = 2y_i + 5$, and the median of x is 25, determine the median of y.

9. In a distribution, with respect to deciles, state which decile represents the Median. 2

10. Find the mean of the observations 1,2,3,…,n when their respective frequencies are $1^2, 2^2, 3^2, \ldots, n^2$. 2

11. In a distribution, what percentage of observations lie below the first quartile (Q_1)? 2

12. The standard deviation of maximum daily temperatures in the Centigrade scale is 3.16. Convert this standard deviation to the Fahrenheit scale and determine its value. 2

13. For the frequency distribution where $x = 1,2,3,\ldots,n$ and the corresponding frequencies are $f_i = 1,2,3,\ldots,n$. find the variance of the distribution. 2

14. If $N = 48$, $\sum f_i y_i = 576$, and the standard deviation is 1, find the value of $\sum f_i x_i^2$. 2

15. Write the mathematical formula for the standard deviation of a frequency distribution. 2

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. An urn contains 4 white and 6 black balls. Two balls are successively drawn from the urn without replacement of the first ball. If the first ball is seen to be white, what is the probability that the 2nd ball is also white? 5

17. A speaks the truth 3 out of 4 times and B 7 times out of 10. They agree in their statement that from a bag containing 6 balls of different colours a white ball has been drawn. Find the probability that the statement is true. 5

18. Draw the ogive (less than and more than type) for the following distribution and use it to find the median: 5

Marks in Math:	20–29	30–39	40–49	50–59	60–69	70–79	80–89	90–99
No. of Students:	7	11	24	32	9	14	2	1

19. Draw the histogram representing the following data: 5
Marks obtained in Mathematics by candidates is given by the following frequency distribution:

Marks:	20–25	25–35	35–45	45–60	60–70	70–75	75–80
No. of candidates:	4	16	18	15	12	5	6

20. Represent the information contained in the following table by a component bar chart 5

Items of expense :	Family A	Family B
Food :	200	300
Clothing :	48	75
Education :	32	40
House rent :	40	75
Misc. :	80	110
Total	400	600

21. Find Q_1 and Q_3 of the following observations: 5

Marks:	10–19	20–29	30–39	40–49	50–59	60–69
No. of Students:	8	11	15	17	12	7

22. The A.M. of the following frequency distribution of marks for a group of 60 students is 30.5. Find the missing values: 5

x:	10	20	—	40	50
f:	8	10	20	15	7

23. The mean weight of a lot of beams is 60 kg. The mean weight of black beams in the lot is 70 kg and that of white beams is 55 kg. Find the proportion of black beams and white beams in the lot. If we have an additional information that there are 150 beams in the lot altogether, then obtain the number of black beams and the number of white beams in the class. 5

24. Find the coefficient of mean deviation about Median for the observations: 2, 5, 6, 9, 13. 5

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. In an examination *30%* of the students failed in Physics, *25%* in Mathematics and *12%* in both Physics and Mathematics. A student is selected at random. Find the probability that: **10**

(i) the student has failed in Physics, if it is known that he has failed in Mathematics.

(ii) the student has failed in at least one of the two subjects.

(iii) the student has passed at least one of the two subjects.

(iv) the student has passed in Mathematics if he failed in Physics.

26. There are two identical urns containing respectively 4 white, 3 red balls and 3 white, 7 red balls. An urn is chosen at random and a ball is drawn from it. Find the probability that the ball is white. If the ball drawn is white, what is the probability that it is from the first urn? **10**

27. Represent the following data using **10**

(i) line chart (ii) bar chart (iii) component bar chart respectively.

Year:	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950
Agricultural: workers (in millions)	6·2	6·9	8·6	9·9	10·9	11·6	11·4	10·5	8·8	6·8
Industrial : workers (in millions)	4·3	6·1	8·8	13·4	18·2	25·4	31·0	38·4	42·9	52·2

28. Draw the Ogive (less than and more than type) for the following distribution **10**

Mid-point:	18	25	32	39	46	53	60	Total
Frequency:	10	15	32	42	26	12	9	146

29. The A.M calculated from the following frequency distribution is known to be 72.5. Find the value of x: **10**

Classes:	30–39	40–49	50–59	60–69	70–79	80–89	90–99
Frequency:	2	3	11	20	x	25	7

30. The frequency distribution of house rent for 30 families in a certain locality is given below: **10**

Rent	1800-2000	2000-2200	2200-2400	2400-2700	2700-3000	3000-3500
No..of families	4	7	10	5	2	2

Find the Median.

31. The mean and standard deviation of marks of 70 students were found to be 65 and 5.2 respectively. Later it was detected that the value 85 was recorded wrongly and therefore it was removed from the data set. Then find the mean and S.D. for the remaining 69 students. **10**

32. The prices of two shares X and Y are given below. Which is more stable? **10**

X: 55 54 52 53 56 58 52 50 51 49
Y: 108 107 105 106 105 107 104 103 101 104

****